Kenny Nguyen

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Southwest Research Institute

Dear Hiring Team,

I am writing to apply for the Computer Scientist / Engineer / Research Engineer - Embedded Software Engineer position with Southwest Research Institute's Space Systems Division. I recently completed my Bachelor of Science in Computer Science at the University of Oregon, and this role stood out to me because it offers the opportunity to contribute to reliable, secure, mission-critical software for space systems.

My background includes software development, C/C++ fundamentals, Python, C#, testing, debugging, automation, and building systems where correctness and maintainability matter. While my strongest project experience so far has been in Python and full-stack development, I am especially interested in growing toward embedded systems, real-time software, and mission-focused engineering. I'm drawn to environments where software has to be carefully designed, tested, and validated because failure has real consequences.

One project that reflects my interest in lower-level software and reliability is MemLock, a static analysis CLI I built for C code using Python and Tree-sitter. The tool detects vulnerability patterns such as buffer overflows, use-after-free risks, format string issues, hardcoded secrets, null pointer risks, and unsafe standard library usage. Building it required me to reason about C source structure, memory-safety risks, parsing, debugging, and how small implementation details can create larger system-level problems.

I have also built larger software systems that required careful design, testing, and troubleshooting. In Backlog, my full-stack job search and workflow platform, I designed database-backed workflows, API integrations, AI-assisted processing, and user-facing application logic using Next.js, TypeScript, Supabase/PostgreSQL, OpenAI/Claude APIs, Playwright, and Vercel. In Syllabify, my university capstone project, I worked on a team to build an academic planning application involving backend APIs, database persistence, authentication, calendar integration, and scheduling logic. These projects strengthened my ability to work through ambiguous requirements, debug across system layers, and build maintainable software with real users in mind.

What interests me most about SwRI is the chance to learn from engineers working on advanced spacecraft avionics, payload processing, flight software, hardware integration, and secure communication systems. I understand that embedded spaceflight software requires a much higher standard of reliability, discipline, and verification than typical application development, and that is exactly what makes the role exciting to me. I would be eager to contribute, learn quickly, and grow within a team building software for demanding space applications.

I am a U.S. citizen, have a valid driver's license, and am able to meet eligibility requirements for a government security investigation. I would be excited for the opportunity to bring my software engineering foundation, debugging mindset, and willingness to learn to Southwest Research Institute.

Thank you for your time and consideration.

Sincerely,

Kenny Nguyen